

Cover page + full report

Linux Lab Week 2, Day 2: Hardening a Linux System

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Part 1: System Updates and Patch Management

LinuxTargetVM4 : Dennis + Nathan, Jason, Dom

- 172.210.68.160
- User: student
- PW: Nathan + Ryan123!

Part 1: System Updates and Patch Management

- 1) Check what updates are available:

```
student@LinuxTargetVM4:~$ whoami
student
student@LinuxTargetVM4:~$ pwd
/home/student
student@LinuxTargetVM4:~$ cd /
student@LinuxTargetVM4:/$ pwd
/
student@LinuxTargetVM4:/$ sudo apt update && apt list --upgradable
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1190 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [282 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1683 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [335 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1424 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [323 kB]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [12.6 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:15 http://azure.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5772 B]
Get:16 http://azure.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12.6 kB]
Get:17 http://azure.archive.ubuntu.com/ubuntu noble-security/main amd64 Packages [930 kB]
Get:18 http://azure.archive.ubuntu.com/ubuntu noble-security/main Translation-en [201 kB]
Get:19 http://azure.archive.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:20 http://azure.archive.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.2 kB]
Get:21 http://azure.archive.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1320 kB]
Get:22 http://azure.archive.ubuntu.com/ubuntu noble-security/restricted Translation-en [302 kB]
Fetched 9092 kB in 2s (3844 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Listing... Done
```

screenshot of the upgradable package list. This is “before” patch evidence.

- 2) Run `sudo apt upgrade -y` to upgrade and check the list using `apt list --upgradable`

```
student@LinuxTargetVM4:/$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
student@LinuxTargetVM4:/$ apt list --upgradable
Listing... Done
student@LinuxTargetVM4:/$
```

Screenshot of upgrade, all upgradable are complete

- 3) Identify if reboot is required

```
student@LinuxTargetVM4:/$ cat /var/run/reboot-required
*** System restart required ***
student@LinuxTargetVM4:/$
```

System reboot IS required, make note and tell instructor/ *Instructor declared no reboot*

- 4) Install unattended-upgrades so future security patches apply automatically:

```
student@LinuxTargetVM4:/$ sudo apt install unattended-upgrades -y
Reading package lists ... Done
Building dependency tree ... Done
Reading state information ... Done
unattended-upgrades is already the newest version (2.9.1+nmu4ubuntu1).
unattended-upgrades set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
student@LinuxTargetVM4:/$
```

Run `sudo dpkg-reconfigure -plow unattended-upgrades` and hit yes. Finally verify it is running.

```
student@LinuxTargetVM4:/$ systemctl status unattended-upgrades
● unattended-upgrades.service - Unattended Upgrades Shutdown
  Loaded: loaded (/usr/lib/systemd/system/unattended-upgrades.service; enabled; preset: enabled)
  Active: active (running) since Wed 2026-08-05 17:09:40 UTC; 6 days ago
    Docs: man:unattended-upgrade(8)
  Main PID: 1131 (unattended-upgr)
    Tasks: 2 (limit: 9434)
  Memory: 9.4M (peak: 11.4M)
    CPU: 164ms
  CGroup: /system.slice/unattended-upgrades.service
          └─1131 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal

Aug 05 17:09:40 LinuxTargetVM4 systemd[1]: Started unattended-upgrades.service - Unattended Upgrades Shutdown.
student@LinuxTargetVM4:/$
```

Part 2: Disabling Unused Services and Ports

Part 2: Disabling Unused Services and Ports

- 1) List all active services: `systemctl list-units --type=service --state=running`

```
student@LinuxTargetVM4:~$ systemctl list-units --type=service --state=running
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
auditd.service                     loaded active running Security Auditing Service
chrony.service                      loaded active running chrony, an NTP client/server
cron.service                        loaded active running Regular background program processing daemon
dbus.service                        loaded active running D-Bus System Message Bus
fail2ban.service                    loaded active running Fail2Ban Service
fwupd.service                       loaded active running Firmware update daemon
getty@tty1.service                  loaded active running Getty on tty1
hv-kvp-daemon.service               loaded active running Hyper-V KVP Protocol Daemon
inetutils-inetd.service             loaded active running GNU Network Utilities internet superserver
multipathd.service                 loaded active running Device-Mapper Multipath Device Controller
networkd-dispatcher.service         loaded active running Dispatcher daemon for systemd-networkd
polkit.service                      loaded active running Authorization Manager
postfix@-.service                   loaded active running Postfix Mail Transport Agent (instance -)
rsyslog.service                     loaded active running System Logging Service
serial-getty@ttyS0.service          loaded active running Serial Getty on ttyS0
ssh.service                         loaded active running OpenBSD Secure Shell server
systemd-journald.service             loaded active running Journal Service
systemd-logind.service              loaded active running User Login Management
systemd-networkd.service            loaded active running Network Configuration
systemd-resolved.service            loaded active running Network Name Resolution
systemd-udevd.service               loaded active running Rule-based Manager for Device Events and Files
udisks2.service                     loaded active running Disk Manager
unattended-upgrades.service          loaded active running Unattended Upgrades Shutdown
user@1000.service                   loaded active running User Manager for UID 1000
wlinuxagent.service                 loaded active running Azure Linux Agent

Legend: LOAD    → Reflects whether the unit definition was properly loaded.
          ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
lines 1-29...skipping...
```

- 2) List all listening ports: `ss -tulpn`

```
student@LinuxTargetVM4:~$ sudo ss -tulpn
Netid      State      Recv-Q     Send-Q     Local Address:Port      Peer Address:Port
Process
udp        UNCONN     0           0           127.0.0.54:53            0.0.0.0:*
users:(("systemd-resolve",pid=141512,fd=16))
udp        UNCONN     0           0           127.0.0.53%lo:53         0.0.0.0:*
users:(("systemd-resolve",pid=141512,fd=14))
udp        UNCONN     0           0           172.16.0.4%eth0:68       0.0.0.0:*
users:(("systemd-network",pid=141344,fd=22))
udp        UNCONN     0           0           127.0.0.1:323            0.0.0.0:*
users:(("chronyd",pid=1124,fd=6))
udp        UNCONN     0           0           [::1]:323                [::]:*
users:(("chronyd",pid=1124,fd=7))
tcp        LISTEN     0           4096        0.0.0.0:22                0.0.0.0:*
users:(("sshd",pid=145285,fd=3),("systemd",pid=1,fd=199))
tcp        LISTEN     0           10         0.0.0.0:23                0.0.0.0:*
users:(("inetutils-inetd",pid=164204,fd=4))
tcp        LISTEN     0           100        0.0.0.0:25                0.0.0.0:*
users:(("master",pid=145632,fd=13))
tcp        LISTEN     0           4096        127.0.0.54:53            0.0.0.0:*
users:(("systemd-resolve",pid=141512,fd=17))
tcp        LISTEN     0           4096        127.0.0.53%lo:53         0.0.0.0:*
users:(("systemd-resolve",pid=141512,fd=15))
tcp        LISTEN     0           4096        [::]:22                  [::]:*
users:(("sshd",pid=145285,fd=4),("systemd",pid=1,fd=200))
tcp        LISTEN     0           100        [::]:25                  [::]:*
users:(("master",pid=145632,fd=14))
```

- 3) For each service/port, determine whether it's actually required for this host's role (banking web/file server).

Note: ssh.service and inetutils-inetd/telnet are kept in order to complete Part 4

- **Postfix.service:** Postfix is providing SMTP, which is why TCP port 25 is listening. Unless this banking web/file server is specifically supposed to act as a mail server, exposing SMTP is unnecessary and increases the attack surface.
- **Multipathd.Service:** This manages multipath storage. On a simple Azure training VM, it is generally unnecessary unless the machine is deliberately configured with multiple paths to the same storage device. We should verify that before disabling it.

Record your reasoning for each one you plan to keep or disable.

- 4) Disable and stop at least two services you determined are unnecessary: systemctl disable --now <service>

Disable Postfix.service:

```
student@LinuxTargetVM4:/$ sudo systemctl disable --now postfix
Synchronizing state of postfix.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable postfix
Removed "/etc/systemd/system/multi-user.target.wants/postfix.service".
student@LinuxTargetVM4:/$ systemctl status postfix.service
o postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; disabled; preset: enabled)
   Active: inactive (dead)
     Docs: man:postfix(1)

Aug 10 18:23:18 LinuxTargetVM4 systemd[1]: Starting postfix.service - Postfix Mail Transport Agent ...
Aug 10 18:23:18 LinuxTargetVM4 systemd[1]: Finished postfix.service - Postfix Mail Transport Agent.
Aug 11 18:46:06 LinuxTargetVM4 systemd[1]: postfix.service: Deactivated successfully.
Aug 11 18:46:06 LinuxTargetVM4 systemd[1]: Stopped postfix.service - Postfix Mail Transport Agent.
student@LinuxTargetVM4:/$
```

Disable Multipathd.service:

```
student@LinuxTargetVM4:~$ sudo systemctl disable --now multipathd.service
Removed "/etc/systemd/system/sysinit.target.wants/multipathd.service".
Removed "/etc/systemd/system/sockets.target.wants/multipathd.socket".
Disabling 'multipathd.service', but its triggering units are still active:
multipathd.socket
Stopping 'multipathd.service', but its triggering units are still active:
multipathd.socket
student@LinuxTargetVM4:~$
```

5) Re-run your port scan and confirm the corresponding ports are no longer listening

```
student@LinuxTargetVM4:~$ sudo ss -tulpn
```

Netid	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port
Process					
udp	UNCONN	0	0	127.0.0.54:53	0.0.0.0:*
users:(("systemd-resolve",pid=141512,fd=16))					
udp	UNCONN	0	0	127.0.0.53%lo:53	0.0.0.0:*
users:(("systemd-resolve",pid=141512,fd=14))					
udp	UNCONN	0	0	172.16.0.4%eth0:68	0.0.0.0:*
users:(("systemd-network",pid=141344,fd=22))					
udp	UNCONN	0	0	127.0.0.1:323	0.0.0.0:*
users:(("chronyd",pid=1124,fd=6))					
udp	UNCONN	0	0	:::1:323	::::*
users:(("chronyd",pid=1124,fd=7))					
tcp	LISTEN	0	4096	0.0.0.0:22	0.0.0.0:*
users:(("sshd",pid=145285,fd=3),("systemd",pid=1,fd=193))					
tcp	LISTEN	0	10	0.0.0.0:23	0.0.0.0:*
users:(("inetutils-inetd",pid=164204,fd=4))					
tcp	LISTEN	0	4096	127.0.0.54:53	0.0.0.0:*
users:(("systemd-resolve",pid=141512,fd=17))					
tcp	LISTEN	0	4096	127.0.0.53%lo:53	0.0.0.0:*
users:(("systemd-resolve",pid=141512,fd=15))					
tcp	LISTEN	0	4096	:::22	:::*
users:(("sshd",pid=145285,fd=4),("systemd",pid=1,fd=194))					

Part 3: Kernel Hardening with sysctl

Part 3: Kernel Hardening with sysctl

1. Review current network-related kernel settings
 - Used sysctl to view the current IPv4 kernel settings.
 - This gave us a baseline before making changes.
 - We checked how the system was currently configured.

```
student@LinuxTargetVM4:~$ pwd
/home/student
student@LinuxTargetVM4:~$ sysctl -a | grep net.ipv4
sysctl: permission denied on key 'kernel.apparmor_display_secid_mode'
sysctl: permission denied on key 'kernel.apparmor_restrict_unprivileged_io_ur
ing'
sysctl: permission denied on key 'kernel.apparmor_restrict_unprivileged_usern
s_complain'
sysctl: permission denied on key 'kernel.apparmor_restrict_unprivileged_usern
s_force'
sysctl: permission denied on key 'kernel.cad_pid'
sysctl: permission denied on key 'kernel.unprivileged_userns_apparmor_policy'
sysctl: permission denied on key 'kernel.usermodehelper.bset'
sysctl: permission denied on key 'kernel.usermodehelper.inheritable'
sysctl: permission denied on key 'net.core.bpf_jit_harden'
sysctl: permission denied on key 'net.core.bpf_jit_kallsyms'
sysctl: permission denied on key 'net.core.bpf_jit_limit'
net.ipv4.cipso_cache_bucket_size = 10
net.ipv4.cipso_cache_enable = 1
net.ipv4.cipso_rbm_optfmt = 0
net.ipv4.cipso_rbm_strictvalid = 1
net.ipv4.conf.all.accept_local = 0
net.ipv4.conf.all.accept_redirects = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.all.arp_accept = 0
net.ipv4.conf.all.arp_announce = 0
net.ipv4.conf.all.arp_evict_nocarrier = 1
net.ipv4.conf.all.arp_filter = 0
net.ipv4.conf.all.arp_ignore = 0
net.ipv4.conf.all.arp_notify = 0
net.ipv4.conf.all.bc_forwarding = 0
net.ipv4.conf.all.bootp_relay = 0
net.ipv4.conf.all.disable_policy = 0
net.ipv4.conf.all.disable_xfrm = 0
net.ipv4.conf.all.drop_gratuitous_arp = 0
net.ipv4.conf.all.drop_unicast_in_l2_multicast = 0
net.ipv4.conf.all.force_igmp_version = 0
```

2. Create the hardening config file
 - Created /etc/sysctl.d/99-hardening.conf.
 - This file stores the security settings we wanted to apply.
 - Putting the settings in a configuration file makes them easier to manage and maintain.

```
student@LinuxTargetVM4:~$ pwd
/home/student
student@LinuxTargetVM4:~$ sudo nano /etc/sysctl.d/99-hardening.conf
student@LinuxTargetVM4:~$
```

```
net.ipv4.ip_forward = 0
net.ipv4.icmp_echo_ignore_broadcasts = 1
net.ipv4.tcp_syncookies = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.default.accept_source_route = 0
```

3. Apply these changes

- Set IP forwarding to 0 so the VM does not act as a router.
- Enabled TCP SYN cookies to help protect against SYN flood attacks.
- Disabled source routing to prevent packets from choosing their own route.
- Configured the system to ignore ICMP broadcast requests.
- These settings reduce unnecessary network exposure.

```
student@LinuxTargetVM4:~$ sudo sysctl --system
* Applying /usr/lib/sysctl.d/10-apparmor.conf ...
* Applying /etc/sysctl.d/10-bufferbloat.conf ...
* Applying /etc/sysctl.d/10-console-messages.conf ...
* Applying /etc/sysctl.d/10-ipv6-privacy.conf ...
* Applying /etc/sysctl.d/10-kernel-hardening.conf ...
* Applying /etc/sysctl.d/10-magic-sysrq.conf ...
* Applying /etc/sysctl.d/10-map-count.conf ...
* Applying /etc/sysctl.d/10-network-security.conf ...
* Applying /etc/sysctl.d/10-ptrace.conf ...
* Applying /etc/sysctl.d/10-zeropage.conf ...
* Applying /usr/lib/sysctl.d/50-pid-max.conf ...
* Applying /etc/sysctl.d/99-cloudimg-ipv6.conf ...
* Applying /etc/sysctl.d/99-cloudimg-udp.conf ...
* Applying /etc/sysctl.d/99-hardening.conf ...
* Applying /usr/lib/sysctl.d/99-protect-links.conf ...
* Applying /etc/sysctl.d/99-sysctl.conf ...
* Applying /etc/sysctl.conf ...
kernel.apparmor_restrict_unprivileged_userns = 1
net.core.default_qdisc = fq_codel
kernel.printk = 4 4 1 7
net.ipv6.conf.all.use_tempaddr = 2
net.ipv6.conf.default.use_tempaddr = 2
kernel.kptr_restrict = 1
kernel.sysrq = 176
vm.max_map_count = 1048576
net.ipv4.conf.default.rp_filter = 2
net.ipv4.conf.all.rp_filter = 2
kernel.yama.ptrace_scope = 1
vm.mmap_min_addr = 65536
kernel.pid_max = 4194304
net.ipv6.conf.all.use_tempaddr = 0
net.ipv6.conf.default.use_tempaddr = 0
net.core.rmem_max = 1048576
net.core.rmem_default = 1048576
net.ipv4.ip_forward = 0
net.ipv4.icmp_echo_ignore_broadcasts = 1
net.ipv4.tcp_syncookies = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.default.accept_source_route = 0
fs.protected_fifos = 1
fs.protected_hardlinks = 1
fs.protected_regular = 2
fs.protected_symlinks = 1
```

4. Verify each setting took effect
- Used sysctl to check the values after applying the changes.
 - Confirmed that each value matched the security settings we configured.
 - This verified that the kernel hardening was successfully applied.
 - The final step makes sure the changes actually took effect rather than just being saved in the file.

```
student@LinuxTargetVM4:~$ sysctl net.ipv4.ip_forward
net.ipv4.ip_forward = 0
student@LinuxTargetVM4:~$ sysctl net.ipv4.icmp_echo_ignore_broadcasts
net.ipv4.icmp_echo_ignore_broadcasts = 1
student@LinuxTargetVM4:~$ sysctl net.ipv4.tcp_syncookies
net.ipv4.tcp_syncookies = 1
student@LinuxTargetVM4:~$ sysctl net.ipv4.conf.all.accept_source_route
net.ipv4.conf.all.accept_source_route = 0 or example:
student@LinuxTargetVM4:~$ sysctl net.ipv4.conf.default.accept_source_route
net.ipv4.conf.default.accept_source_route = 0
student@LinuxTargetVM4:~$ █
```

Part 4: Mandatory Access Control with AppArmor

Part 4: Mandatory Access Control with AppArmor

- 1) Check AppArmor's current status and record which profiles are in enforce mode, complain mode, unloaded.

```
student@LinuxTargetVM4:~$ sudo aa-status
apparmor module is loaded.
118 profiles are loaded.
```

```
24 profiles are in enforce mode.
/usr/bin/man
/usr/lib/snapd/snap-confine
/usr/sbin/chronyd
lsb_release
man_filter
man_groff
nvidia_modprobe
nvidia_modprobe//kmod
plasmashell
plasmashell//QtWebEngineProcess
rsyslogd
tcpdump
ubuntu_pro_apt_news
ubuntu_pro_esm_cache
ubuntu_pro_esm_cache//apt_methods
ubuntu_pro_esm_cache//apt_methods_gpgv
ubuntu_pro_esm_cache//cloud_id
ubuntu_pro_esm_cache//dpkg
ubuntu_pro_esm_cache//ps
ubuntu_pro_esm_cache//ubuntu_distro_info
ubuntu_pro_esm_cache_systemctl
ubuntu_pro_esm_cache_systemd_detect_virt
unix-chkpwd
unprivileged usersns
```

(24 profiles in enforce mode)

```
4 profiles are in complain mode.
transmission-cli
transmission-daemon
transmission-gtk
transmission-qt
```

(4 profiles in enforce mode)

```
90 profiles are in unconfined mode.
1password
Discord
MongoDB Compass
QtWebEngineProcess
balena-etcher
brave
buildah
cam
ch-checkns
ch-run
chrome
crun
devhelp
element-desktop
epiphany
evolution
firefox
flatpak
foliate
geary
github-desktop
goldendict
ipa_verify
kchmviewer
keybase
lc-compliance
libcamerify
linux-sandbox
loupe
lxc-attach
lxc-create
lxc-destroy
lxc-execute
lxc-stop
lxc-unshare
lxc-usernsexec
mmdesktop
msedge
nautilus
notepadqq
obsidian
opam
opera
pageedit
podman
polypane
privacybrowser
qcam
qmapshack
qutebrowser
rootlesskit
rpm
rssguard
runc
sbuid
sbuid-abort
sbuid-adduser
sbuid-apt
sbuid-checkpackages
sbuid-clean
sbuid-createroot
sbuid-destroyroot
sbuid-distupgrade
sbuid-hold
sbuid-shell
sbuid-unhold
sbuid-update
sbuid-upgrade
scide
signal-desktop
slack
slirp/netns
```

```
steam
stress-ng
surfshark
systemd-coredump
thunderbird
toybox
trinity
tup
tuxedo-control-center
userbindmount
uwsgi-core
vden
virtiofsd
vivaldi-bin
vpnn
vscode
wike
wpcorn
```

(90 profiles in unconfined mode)

- 2) Choose one running service without an active profile and put it into enforce mode

```
student@LinuxTargetVM4:~$ sudo aa-genprof /usr/sbin/telnetd
Updating AppArmor profiles in /etc/apparmor.d.
Writing updated profile for /usr/sbin/telnetd.
Setting /usr/sbin/telnetd to complain mode.

Before you begin, you may wish to check if a
profile already exists for the application you
wish to confine. See the following wiki page for
more information:
https://gitlab.com/apparmor/apparmor/wikis/Profiles

Profiling: /usr/sbin/telnetd

Please start the application to be profiled in
another window and exercise its functionality now.

Once completed, select the "Scan" option below in
order to scan the system logs for AppArmor events.

For each AppArmor event, you will be given the
opportunity to choose whether the access should be
allowed or denied.

[(S)can system log for AppArmor events] / (F)inish
```

- 3) Restart the affected service and confirm it still functions correctly under enforcement
- 4) Check the AppArmor logs for any denials, and decide whether each denial is expected behavior being correctly blocked, or a legitimate function the needs a profile adjustment.

Hardening Sum. & Reflection Questions

Deliverables

- Your patch status before/after (Part 1) including confirmation of unattended-upgrades
- Your service/port inventory with keep-or-disable reasoning, and confirmation of the ports you closed
- Your /etc/sysctl.d/99-hardening.conf file and verification of each applied parameter
- Your AppArmor before/after status and any denial log entries you reviewed
- Your written hardening summary

Hardening Summary

The Linux host was brought up to date by installing all available package updates, including any kernel-related updates discovered by apt, and unattended upgrades were enabled for future security patches. During the service assessment, Postfix was stopped because the server did not require SMTP capability, hence removing the unwanted exposure of TCP port 25. Multipathd was also deactivated since the host did not use multipath storage and so did not require the service to run. Kernel networking settings were hardened by turning off IP forwarding and source routing, enabling TCP SYN cookies, and instructing the system to ignore ICMP broadcast requests. These modifications minimize susceptibility to needless network routing and numerous prevalent network-based threats.

Reflection Questions

1. Which change today felt riskiest to make on a production-bound host, and how did you validate it didn't break anything?

The riskiest change was enabling the new AppArmor profile for Telnet in enforce mode, because an incorrect profile could block legitimate activity or prevent the service from functioning. I confirmed the modification by restarting inetutils-inetd, connected via Telnet, performing standard commands like whoami and ls, and then reviewing the AppArmor logs for any unusual denials. Telnet continued to operate regularly, so I knew the profile had not disrupted the service.

2. Why might disabling a service be a better fix than just blocking its port with a firewall rule? Are there cases where you'd want both?

Disabling a service might be a better fix as it kills the running process entirely. This frees up system memory and CPU power but it also reduces risk by preventing local exploitation. A firewall rule only filters network traffic hitting that port, meaning an attacker with local access could still interact with and exploit a vulnerable running service. A case where both should be utilized is in defense in depth models where you disable services you don't use and configure firewall rules to restrict access on required services.

3. How does mandatory access control (AppArmor) protect a host differently than the least-privilege user/file permissions work from Week 1?

The Discretionary Access Control (DAC) we used in Week 1 relies on user and group file permissions, which grant any process running as a user full access to all files that user owns. In this lab, Mandatory Access Control (MAC), utilizing AppArmor, uses profiles to dictate which files and network calls that an individual program may access regardless of user privileges.